



TB AUDIO PLUG-INS  
Team Impulse Impact

## USER MANUAL

# TB Resonator

## Tunable Modal Resonator

A playable modal instrument that turns any input into a tuned physical body, with eight resonant modes, zero latency, and tails up to 30 seconds.

8 resonant modes

Tune C1 to C7

Memory up to 30 s

Zero latency



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## 01 Overview

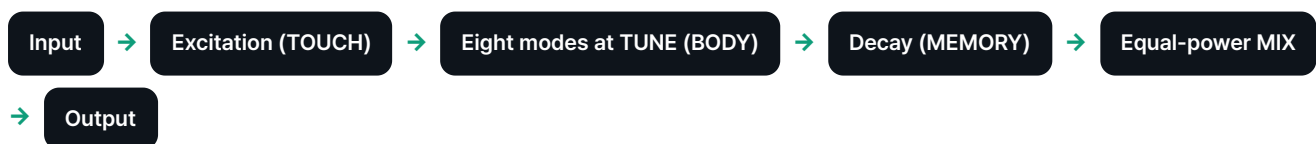
TB Resonator turns any input into a tuned physical object. Eight stable resonant modes give clicks, drums, noise, guitars, pads, or vocals a defined pitch, material, and memory.

**TUNE** sets the musical centre. **BODY** moves the material between harmonic and inharmonic structures. **TOUCH** changes how transients and sustained energy excite the object. **MEMORY** controls how long the modes ring, and **MIX** balances the transformation with the original. TB Resonator adds no latency, works in mono or stereo, and its tails can last up to 30 seconds.

### Key features

- Eight resonant modes tuned from C1 to C7
- BODY from SOFT through WOOD, STRING, and GLASS to METAL
- TOUCH from STRIKE through PLUCK, NATURAL, and OPEN to DRIVE
- MEMORY from 0.04 to 30 seconds
- Equal-power MIX
- Zero latency, mono or stereo
- 12 factory presets in three groups, user presets, undo, and redo

### Signal flow



#### About this manual

This manual describes TB Resonator 1.0.1 for Windows (VST3). The current macOS build is also version 1.0.1; small differences between platforms are possible.

## 02 Quick start

- 1 Insert TB Resonator on drums, a guitar, a pad, or a vocal. The defaults are **TUNE** C3, **TOUCH** 50%, **BODY** 50%, **MEMORY** 1.10 s, and **MIX** 35%.
- 2 Set **TUNE** **2** to the key of the song, or to a note that fits the chord.

- 3 Choose the material with **BODY** 4 : soft and woody to the left, glassy and metallic to the right.
- 4 Set how the input strikes the object with **TOUCH** 3 .
- 5 Set the ring length with **MEMORY** 5 and the blend with **MIX** 6 .

## 03 Interface



TB Resonator 1.0 interface.

- 1 **Preset bar** — previous and next factory preset (hold to browse), the preset list shown as group / name, Save, Undo, and Redo.
- 2 **TUNE** — the resonant note with its frequency, from C1 to C7.
- 3 **TOUCH** — excitation, from STRIKE to DRIVE.
- 4 **BODY** — material, from SOFT to METAL.
- 5 **MEMORY** — ring time, from SHORT to LONG.
- 6 **MIX** — from DRY to WET.

## 04 Controls

| Control       | Range · default                          | What it does                                                                                                                                                 |
|---------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TUNE</b>   | C1 to C7 (MIDI 24 to 96) · C3 (130.8 Hz) | Chooses the fundamental note remembered by the resonator. The display shows the note and its frequency.                                                      |
| <b>TOUCH</b>  | 0 to 100% · 50% (NATURAL)                | Moves from transient-only excitation to open, driven excitation. The readout names the range: STRIKE, PLUCK, NATURAL, OPEN, DRIVE.                           |
| <b>BODY</b>   | 0 to 100% · 50% (STRING)                 | Morphs spacing, brightness, and high-mode decay as one material, from harmonic to inharmonic. The readout names the range: SOFT, WOOD, STRING, GLASS, METAL. |
| <b>MEMORY</b> | 0.04 to 30 s · 1.10 s                    | Sets how long the resonant object remembers each excitation.                                                                                                 |
| <b>MIX</b>    | 0 to 100% · 35%                          | Blends the resonant object with the original signal at equal power.                                                                                          |

TB Resonator also has a host Bypass parameter for comparing the dry and processed sound.

Long MEMORY settings and strong low notes can build up level and low-frequency energy. Watch the output level and low end when you lengthen the ring.

## 05 Presets

Step through factory presets with the arrows; hold an arrow to browse continuously. Open the list to choose a factory preset or load a user preset. Save stores the current settings as a user preset. Undo and Redo step through recent parameter changes.

| Group   | Preset             | TUNE | TOUCH | BODY | MEMORY | MIX |
|---------|--------------------|------|-------|------|--------|-----|
| Start   | Init               | C3   | 50%   | 50%  | 1.10 s | 35% |
| Start   | Soft Chamber       | G3   | 38%   | 18%  | 0.85 s | 28% |
| Start   | Open String        | A2   | 55%   | 50%  | 2.4 s  | 42% |
| Start   | Short Body         | G2   | 20%   | 28%  | 0.14 s | 45% |
| Source  | Tuned Drum         | C2   | 16%   | 12%  | 0.65 s | 55% |
| Source  | Sympathetic Guitar | E2   | 42%   | 42%  | 3.8 s  | 48% |
| Source  | Vocal Glass        | C4   | 62%   | 76%  | 1.65 s | 52% |
| Source  | Bass Wood          | E1   | 58%   | 8%   | 1.25 s | 50% |
| Texture | Harmonic Bloom     | G3   | 68%   | 50%  | 8 s    | 70% |
| Texture | Metallic Bloom     | C4   | 72%   | 90%  | 10.5 s | 75% |
| Texture | Struck Chime       | C5   | 8%    | 94%  | 5.2 s  | 62% |
| Texture | Driven Drone       | A1   | 96%   | 66%  | 24 s   | 85% |

## 06 Recipes

Starting points for common jobs. Adjust them by ear.

### Tuned drums

Give a kick, tom, or percussion loop a pitch.

|        |                 |
|--------|-----------------|
| Preset | Tuned Drum      |
| TUNE   | The song's root |
| MEMORY | 0.3 to 1 s      |

- 1 Keep TOUCH low so only the hits excite the body.

### Sympathetic strings

A ringing halo around guitar or piano.

|        |                    |
|--------|--------------------|
| Preset | Sympathetic Guitar |
| BODY   | STRING             |
| MIX    | 25 to 45%          |

- 1 Tune to a note that sits in every chord of the section.

### Glass voice

A tuned, glassy layer on vocals.

Preset                      Vocal Glass

BODY                      GLASS

1 Lower MIX if the words lose clarity.

### Drone bed

A sustained, driven texture from any input.

Preset                      Driven Drone

TOUCH                      DRIVE

MEMORY                      15 to 30 s

1 Watch the level as the tail builds up.

## 07 Tips and troubleshooting

| Symptom                                 | What to check                                              |
|-----------------------------------------|------------------------------------------------------------|
| The resonance clashes with the music.   | Retune TUNE to the key, or to a note shared by the chords. |
| The sound gets boomy or too loud.       | Shorten MEMORY, raise TUNE, or lower MIX.                  |
| Only hits ring, sustained parts do not. | Raise TOUCH toward OPEN or DRIVE.                          |
| It sounds too metallic.                 | Move BODY toward WOOD or STRING.                           |
| Nothing changes.                        | MIX is at 0%, or the host Bypass parameter is on.          |

## 08 Parameter reference

| Panel label | Host parameter | Range                         | Default |
|-------------|----------------|-------------------------------|---------|
| TUNE        | Tune           | MIDI note 24 to 96 (C1 to C7) | 48 (C3) |
| TOUCH       | Touch          | 0 to 1 (0 to 100%)            | 0.5     |
| BODY        | Body           | 0 to 1 (0 to 100%)            | 0.5     |
| MEMORY      | Memory         | 0.04 to 30 s                  | 1.10 s  |
| MIX         | Mix            | 0 to 1 (0 to 100%)            | 0.35    |
| —           | Bypass         | Off, On                       | Off     |

## 09 Version history

| Version            | Changes                                                                                                                                                                                  |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.0.1 · 2026-08-21 | Brought the current Tune, Body, Memory, Touch, and Mix design and the latest resonance processing build to Windows VST3, and aligned the catalog, VST3, and installer versions at 1.0.1. |
| 1.0.0 · 2026-08-02 | First release: Tune, Body, Memory, Touch, and Mix, with 12 factory presets and user presets.                                                                                             |

## 10 Support and license

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Report problems or request features at [github.com/TaebeumKim/TB\\_Audio-Plugins/issues](https://github.com/TaebeumKim/TB_Audio-Plugins/issues). Include the plug-in version, your DAW, and your operating system.

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